

Jacob Helwig

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Education

Texas A&M University, Ph.D. Computer Science

August 2021-May 2026

The University of Texas at Austin, BS Mathematics

August 2016-May 2021

Work Experience

LinkedIn, CoreAI Intern

Summer 2024, Summer 2025

- **Summer 2025:** Pretrained transformers for sequential recommendation. Key challenges include correct weight and optimizer state updates with sparse gradients as a result of high percentage of unused embeddings due to large item cardinality/vocab size.
- **Summer 2024:** Developed full pipeline for large-scale training of causal transformers for sequential recommendation. Key challenges include memory management due to large cardinality/vocab size leading to large model head and logits vector. Solved using kernel fusion of model head with cross-entropy loss.

Los Alamos National Laboratory, Applied ML Fellow

Summer 2023

- Project: Solving the Elastic Wave Equation Using Deep Learning

OSS Contributions

verl, Volcano Engine Reinforcement Learning for LLMs ([All Merged PRs](#))

- **On-Policy Distillation:** vLLM teacher server supporting LLM/VLM distillation with token-in/token-out. *Trainer backends:* FSDP/Megatron. *Distillation losses:* KL estimator/top- k KL with Megatron vocab parallel autograd function. *Optimization algorithm:* policy gradient with negative KL reward/backpropagate distillation loss.
- **Multi-Teacher On-Policy Distillation:** Domain-specific teachers, as used by DeepSeek V4, GLM-5, MiMo-V2-Flash.

Publications and Talks

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|----------------|---|
| [ICLR 2026] | A Two-Phase Deep Learning Framework for Adaptive Time-Stepping in High-Speed Flow Modeling
First author, paper , code |
| [NeurIPS 2024] | A Geometry-Aware Message Passing Neural Network for Modeling Aerodynamics over Airfoils (talk at the NeurIPS 2024 ML4CFD Competition)
Best student solution, paper , code |
| [ICML 2024] | Equivariance via Minimal Frame Averaging for More Symmetries and Efficiency
Spotlight, paper |
| [FnT ML] | Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems
Co-first author, paper |
| [ICLR 2024] | SineNet: Learning Temporal Dynamics in Time-Dependent PDEs
Co-first author, paper , code |
| [LoG 2023] | High-fidelity Fluid Flow Reconstruction
Paper |
| [ICML 2023] | Group Equivariant Fourier Neural Operators for Partial Differential Equations
Co-first author, paper , code |
| [ACM TOMS] | A Covariate-Dependent Approach to Gaussian Graphical Modeling in R
First author, paper , R package |
| [Preprint] | An Approximate Bayesian Approach to Covariate-dependent Graphical Modeling
Paper |

Honors

Ruth J. & Howard F. Newton Memorial Graduate Student Teaching Award In Statistics, 2022 Recipient

- “Jacob was selected to receive the Newton Teaching Award based on his outstanding evaluations by the instructors he was Teaching Assistant for during the Fall 2021 and Spring 2022 semesters.” ([award details](#))

UT Austin University Honors, Fall 2019, Spring 2020, Fall 2020, & Spring 2021

- “To be included, a student must earn at least 45 grade points [and] a grade point average of at least 3.50”